



Uncertainty and the uncovered interest parity condition: How are they related?

Nelson R. Ramírez-Rondán¹ · Marco E. Terrones²

Accepted: 9 April 2024 / Published online: 23 May 2024

© The Author(s) under exclusive licence to Kiel Institute for the World Economy 2024

Abstract

A well-established literature in open macroeconomics documents the failure of the uncovered interest parity (UIP) condition. While a host of factors have been examined as possible reasons for this failure, the role of uncertainty is not fully understood. This paper examines how economic policy uncertainty affects the UIP condition in a panel of fourteen economies over 2003:1–2021:12. There is evidence of an uncertainty threshold on the UIP condition under risk neutrality. This threshold splits the sample endogenously into two different regimes—a “low-uncertainty regime” and a “high-uncertainty regime.” While the UIP condition holds in the former regime, it fails in the latter. This finding is robust to the inclusion of risk-premium proxies to allow for risk aversion, *ex-post* exchange rates, different deposit maturities, and estimation methods.

Keywords Exchange rates · Uncertainty · Panel threshold models

JEL Classification F31 · E43 · D80

1 Introduction

Over the past four decades, the validity of the uncovered interest parity (UIP) condition has been extensively examined but with mostly unfavorable results. The UIP condition establishes that the expected exchange rate change should equal the interest differential between the assets of two countries, domestic and foreign. This condition has been examined by regressing the (expected) change in the bilateral exchange rate (domestic currency in units of the foreign currency) on the interest rate differential between interest-bearing domestic and foreign assets, the so-called

✉ Nelson R. Ramírez-Rondán
nramron@gmail.com

¹ CEMLA, Durango 54, Mexico City 06700, Mexico

² Universidad del Pacífico, Lima, Peru

Fama regression.^{1,2} In this regression, the coefficient of the interest rate differential or slope coefficient implied by the UIP condition is equal to one. Most empirical studies, however, report that the estimated slope coefficient is either negative or smaller than one (Burnside et al., 2006; Froot and Thaler, 1990; among others). This result, labeled the UIP puzzle, implies that the expected returns across currencies do not equalize, as economies with a higher interest rate display a more appreciated exchange rate than that indicated by the UIP condition.³

Several factors have been identified as possible reasons for this puzzle. First, most empirical studies have examined the joint hypothesis of UIP and rational expectations (Bussière et al., 2022; Chinn & Meredith, 2004). Under this joint null hypothesis, the slope coefficient must equal one. A fundamental assumption behind the Fama regression is that the covered interest parity (CIP) condition holds. According to this condition, the return on a domestic asset and the return on a foreign asset expressed in domestic currency using the forward exchange rate must be equal. A vast literature documents that CIP generally held in the data until the Global Financial Crisis (GFC).⁴ Since then, some studies have reported deviations from the CIP condition (Cerutti et al., 2021; Du et al., 2018). These deviations seem to reflect a combination of structural factors, such as post-crisis changes in financial regulation, and transitory factors, such as divergent monetary policies across countries and the 2016 reform of the U.S. prime money market fund (Cerutti et al., 2021). Other studies have examined whether exchange rate expectations are rational, that is whether exchange rate forecasts are unbiased. There is ample evidence that suggests the failure of the unbiased hypothesis (Flood and Rose, 2002; Froot and Thaler, 1990; Meese and Rogoff, 1983; among others) since prediction errors are often negatively related to the interest differential so that estimated slopes are negative.

Second, some studies have noted that the Fama regression equation may be omitting one or more important explanatory variables with potential downward bias in the slope coefficient. If economic agents are risk-averse, then the Fama regression equation needs to include a time-varying risk premia term. Part of the literature suggests that in the case of standard consumer preferences, with a risk aversion coefficient based on empirical studies, risk premiums might not vary sufficiently to generate a negative slope coefficient; thus, the relevance of this variable is downplayed (Engel, 1996). More recent studies, however, have shown that by using non-standard preferences (Lustig et al.,

¹ The expected change in the spot exchange rate refers to the h -period change. A positive change in the exchange rate means that the domestic currency has depreciated. The interest rate differential refers to the difference between the domestic and foreign currency deposit rates, with a maturity of h -periods.

² Fama (1984) established a relationship between the expected exchange rate change and the difference between the forward value of the exchange rate and its spot rate. Several simplifying assumptions are then needed to transform this relationship into the UIP regression equation, including holding the covered interest rate parity, rational expectations, risk neutrality, etc.

³ Several studies have shown that this puzzle is more prevalent when using short-term (monthly, quarterly) data. When using long-term (5-years, 10-years) data, the estimated slope coefficients are positive and closer to one (Chinn & Meredith, 2004; Chinn & Quayyum, 2013).

⁴ During this period there were small and transient departure episodes from the CIP condition (see, for instance, Akram et al., 2008) often associated with financial uncertainty and market turbulence. Despite these episodes, assuming that the CIP condition held before the GFC was safe.

2011; Verdelhan, 2010) it is possible to arrive at a time-varying risk premia term that has the potential to explain the UIP puzzle. Some variables that could affect this time-varying term are uncertainty, business cycle, capital controls, exchange rate regime, inflation rate, and terms of trade (Colacito et al., 2020; Farhi & Werning, 2014; Lustig et al., 2019). Nevertheless, these studies are unable to match other essential characteristics of the data, such as the evolution of the real exchange rate (Engel, 2016).

Third, other studies have explored the presence of non-linearities in the Fama regression equation. The presence of non-linearities was initially justified as the result of transaction costs (Hollifield & Uppal, 1997) or limits on speculation (Sarno et al., 2006). The Fama equation is, in this case, modified by a smooth transition function that establishes how the deviations from UIP converge toward zero. The transition function is determined by the Sharpe ratio (Sarno et al., 2006), the forward premium (Baillie & Kilic, 2006), or other variables related to foreign exchange traders' opportunity costs. It is assumed that when these opportunity costs are high enough, traders carry out exchange transactions that make UIP possible. Threshold autoregressive models have also been used by Balke and Wohar (1998) and Peel and Taylor (2002) to examine deviations from CIP. More recently, Farhi and Gabaix (2016) developed a disaster risk model with two regimes—"disaster" and "non-disaster"—where the lumping of data across these regimes leads to a Fama slope estimate of less than one, i.e., failure of the UIP parity condition (see also Corsetti and Marin, 2020).

Uncertainty could mainly affect the Fama regression equation through its asymmetric effects on financial markets. Heightened uncertainty is often accompanied by financial market instability and the dissipation of two-way healthy market operations. Credit conditions tighten, arbitrage opportunity gains become more uncertain, and asset trade volumes fall (Bordo et al., 2016; Ismailov & Rossi, 2018; Rehse et al., 2019). This partly reflects that increased uncertainty is related to rising transaction costs, including higher information acquisition and supervision costs. Lastly, heightened uncertainty is also associated with higher excess returns in currency carry trade operations (Berg & Mark, 2018; Husted et al., 2018).

Because of these subtle effects, we postulate that uncertainty affects the Fama equation non-linearly by splitting the sample following a threshold panel regression model. This parsimonious way of introducing uncertainty into the UIP analysis allows for differences in the regression parameters across uncertainty regimes (Hansen, 2000).⁵ To this end, we utilize consensus forecast survey data on exchange rates and news-based economic policy uncertainty indices in the empirical analysis. The adequacy of exchange rate survey data for empirical work is documented in Stavrakeva and Tang (2015). Furthermore, using the economic policy uncertainty index (Baker et al., 2016) as a proxy of uncertainty is largely justified because of its country coverage. Our baseline is thus the 3-month Fama panel regression for 14 economies. By exploiting the variation across countries and over time, panel data

⁵ Note that, as a reviewer highlighted, the disaster event represents extreme events, which is not the case with high and low uncertainty regimes. In our approach, the formulation allows for two event states of "low" and "high" uncertainty, where one event is anticipated with a high probability and the other with a low probability (difficult to predict).

allows for greater model flexibility, efficiency gains, and control for the effects of omitted variables.⁶

We find that uncertainty is a threshold variable in the Fama regression that splits the sample into two regimes which we denote the “low-uncertainty” and “high-uncertainty” regimes, because of their lower or higher than threshold levels of uncertainty. In the low-uncertainty regime, the slope coefficient estimate of the Fama regression is not significantly different from one, which means that the UIP condition holds. In contrast, in the high-uncertainty regime, the slope coefficient is significantly different from one, so the UIP condition does not hold. This means that carry trade is profitable; however, carry trade activities are likely to be limited by the liquidity squeeze that often characterizes this high-uncertainty regime. These results are robust to the inclusion of other regressors, changes in term maturity, proxies of uncertainty, and estimation methods.

Our findings are related to those reported by Ismailov and Rossi (2018). Based on single-country Fama regressions for five industrialized economies over the period 1993:11–2015:1, they find that the UIP tends to hold in low-uncertainty environments and fail in high-uncertainty ones. However, separating these two uncertainty environments follows an *ad hoc* statistical rule applied before the regression analysis. In particular, the high-uncertainty environments are those in the top quartile of each country’s exchange rate uncertainty index, while the low-uncertainty ones are the rest. Our study extends theirs because it uses a more general non-linear framework and a more comprehensive dataset, where we rely on more widely used uncertainty measures and survey-based exchange rate expectations.⁷ Lastly, we do not *a priori* impose the existence of two uncertainty environments, as the data determines the number of uncertainty regimes endogenously.

How sensitive are our main findings to omitting a risk premium term or using survey-based exchange rate expectations? The inclusion of a risk premium term is the result of having risk-averse investors in frameworks such as that of Corsetti and Marin (2020) (see also Lloyd and Marin, 2020) as the covariance between the stochastic discount factor and the change in the nominal exchange rate is non-zero. Given that there are no appropriate risk premium measures, we utilize several proxies from the literature, including uncertainty and relative term slope. Importantly, when using uncertainty both as a threshold variable and as an additional regressor both terms are significant, although the latter term is numerically unimportant in the relevant cases.⁸ Our main results also survive in all other cases, suggesting that uncertainty is a crucial threshold variable affecting the UIP condition.

Next, we examine whether using survey-based exchange rate expectations drives our main results. To evaluate this possibility, we re-estimate the threshold Fama regression equation using *ex-post* exchange rates. As in the case of survey-based

⁶ Recent exchange rate studies (Colacito et al., 2020; Herger, 2016) have shown the usefulness of these cross section and panel methods.

⁷ Ismailov and Rossi (2018) use an exchange rate uncertainty measure that is constructed by comparing the realized forecast error of the exchange rate with the unconditional forecast error distribution of the same variable. Their uncertainty measure, however, depends on the Fama regression model used to forecast exchange rates.

⁸ In any case, recent work highlights the importance of distinguishing between uncertainty and risk aversion shocks, which are not the same and have different real and financial effects (Berthold, 2023).

exchange rate expectations, we find that the UIP holds during the low-uncertainty regime but fails in the high-uncertainty regime. This finding is reassuring as we can conclude that our main results do not depend on the failure of rational expectations. The remainder of this paper is organized as follows. In Sect. 2, we discuss the methodology and dataset we utilize in this study. In Sect. 3, we examine the Fama regression model using a panel threshold framework where economic policy uncertainty is the threshold variable. In Sect. 4, we conclude with a summary of our findings. Finally, in the Appendix A, we discuss certain robustness exercises.

2 Methodology and data

In this section, we briefly discuss our methodology and dataset. As noted earlier, we embed the Fama equation within a panel threshold regression model, whereby uncertainty is the threshold variable that splits the sample into two or more regimes. The dataset comprises monthly information for 14 countries over the period 2003:1–2021:12. The data comes from several sources, including Bloomberg, Datastream, Economic Policy Uncertainty, and Consensus Forecast.

2.1 The Fama equation

The uncovered interest parity condition states that in a world with perfect capital mobility and risk-neutral investors, the returns in domestic currency of an asset denominated in both domestic and foreign currency must be equal. Thus, Eq. (1) should hold,

$$(1 + i_{h,t}) = \left(1 + i_{h,t}^*\right) \frac{E_t S_{t+h}}{S_t}, \quad (1)$$

where $i_{h,t}$ and $i_{h,t}^*$ are the domestic and foreign interest rates for a fixed asset that matures in h periods. S_t is the spot exchange rate (domestic currency/foreign currency) in period t , E_t is the expectation conditional on information in period t , and S_{t+h} is the spot exchange rate in period $t + h$. Taking the logarithms of both sides of the equation and ignoring Jensen's inequality, we obtain the following linear equation, which is known as the Fama regression equation

$$E_t s_{t+h} - s_t = i_{h,t} - i_{h,t}^*, \quad (2)$$

where s_{t+h} is the log spot exchange rate in $t + h$.

To estimate the Fama regression equation, we need a proxy for exchange rate expectations. In the empirical literature, two such proxies have been used. The most common one is the *ex-post* exchange rate expectation. Proceeding this way, the null is a joint hypothesis of UIP and rational expectations. However, as noted earlier, there is ample evidence to suggest that rational expectations do not hold. This has prompted several researchers to use survey-based exchange rate expectations. For instance, Bussière et al. (2022), and Chinn and Frankel (2019) find that when using

survey-based data, the deviations from UIP are much smaller than those resulting from *ex-post* expectations. In this paper, we utilize the survey-based exchange rate expectations obtained from the Consensus Forecast.⁹

The *ex-ante* Fama regression equation for a given country is now given by

$$\hat{s}_{t+h} - s_t = \alpha + \beta(i_{h,t} - i_{h,t}^*) + e_{t+h}, \quad (3)$$

where $\hat{s}_{t+h}$ is the survey-based exchange rate expectation and e_{t+h} is an error term. When the survey-based exchange rate expectations are not biased, i.e., they have mean zero, then e_{t+h} also has mean zero. From now on, we will assume that this is the case. When the intercept α is equal to zero and the slope β is equal to one, then the UIP hypothesis holds in the data. In more realistic settings, however, the intercept might be different from zero; for instance, when there are differentiated and time-invariant taxes to the domestic and foreign currency deposits, constant risk aversion, etc.

The *ex-ante* Fama panel regression equation is given by

$$\hat{s}_{it+h} - s_{it} = \mu_i + \alpha + \beta(i_{h,it} - i_{h,it}^*) + e_{it+h}, \quad (4)$$

where the sub-index i refers to the country i , $1 \leq i \leq n$, and μ_i is an unobserved country-specific effect which is assumed to be fixed. The *ex-ante* Fama panel regression model allows us to control for country-unobservable characteristics persistent over time and to capture common parameters, such as the slope coefficient.

2.2 The threshold Fama panel regression model

We initially postulate that uncertainty affects the *ex-ante* Fama panel regression by splitting the sample into two regimes. This can be interpreted as a reduced-form formulation of the equilibrium condition that requires that the expected discounted cash flow from the arbitrage position in two-currency investments for two states of nature—say a “high-uncertainty” regime and a “low-uncertainty” regime—equals zero. For the sake of simplicity assume that the probability of being in either of these regimes is determined by the level of uncertainty relative to an uncertainty threshold γ . Thus, the resulting *ex-ante* threshold Fama panel regression model can be written as

$$\hat{s}_{it+h} - s_{it} = \mu_i + (\alpha_1 + \beta_1(i_{h,it} - i_{h,it}^*))1(q_{it} \leq \gamma) + (\alpha_2 + \beta_2(i_{h,it} - i_{h,it}^*))1(q_{it} > \gamma) + e_{it+h}, \quad (5)$$

where q_{it} stands for uncertainty in country i and period t (threshold variable), γ is the threshold parameter which needs to be estimated along with the other parameters in the regression equation, and $1(\cdot)$ is the indicator function, which takes on the value of 1 if the inequality inside the bracket is true, and 0 otherwise. In this model, the marginal effect in the expected exchange rate due to a shock in the interest rate differentials is given by

⁹ There is evidence that this survey data does not follow mechanical rules associated with either the forward exchange rate or interest rate differentials (Stavrakeva & Tang, 2015).

$$\frac{\partial(\hat{s}_{it+h} - s_{it})}{\partial(i_{h,it} - i_{h,it}^*)} = \begin{cases} \beta_1 & \text{if } q_{it} < \gamma \\ \beta_2 & \text{if } q_{it} \geq \gamma, \end{cases} \quad (6)$$

where β_1 could be different from β_2 ; that is, the slopes could differ across regimes.

The empirical analysis of these models involves three steps: estimation, inference, and testing. The estimation and inference theory for these models is developed in Hansen (2000). In particular, he proposes a method to construct confidence intervals for the threshold parameter, γ , in a simple closed-form expression. Hansen (1999) extends these methods to a balanced panel data context. After estimating Eq. (5), we need to test whether the threshold parameter is statistically significant, whether $\alpha_1 = \alpha_2$ and $\beta_1 = \beta_2$ which is the hypothesis of no threshold effect, and whether each $\beta_j = 1$ for $j=1,2$, which is the hypothesis of UIP. We expect the deviations from UIP to be smaller in the low uncertainty regime, when $q_{it} < \gamma$, than in the high uncertainty regime, when $q_{it} \geq \gamma$. Next, we need to determine the number of uncertainty thresholds in this model. To this end, we perform a series of sequential tests—a test of no threshold against one threshold, a test of one threshold versus two thresholds, and so on. We can proceed this way because the regimes in this framework are observable *ex-post*.

Our empirical framework is more general than the one used by Ismailov and Rossi (2018). First, in contrast to the exogenous sample split following *ad hoc* rules as done by these authors, we apply the sample split jointly to the estimation of the Fama regression parameters. Second, the number of regimes in which the sample could be split in our framework could exceed two—as dictated by the sample. Third, we are using a panel regression with fixed effects, which should deliver better results than country-specific regression models given the short country samples. Lastly, we estimate an *ex-ante* Fama regression model as opposed to an *ex-post* Fama regression model.

2.2.1 Parameter estimation

The within transformation suggested by Hansen (1999) is given by

$$\begin{aligned} (\hat{s}_{it+h} - s_{it})^+ &= (\alpha_1 - \alpha_2)1(q_{it} \leq \gamma)^+ + \beta_1(i_{h,it} - i_{h,it}^*)^+ \\ &\quad + \beta_2(i_{h,it} - i_{h,it}^*)^\pm + e_{it+h}^+, \end{aligned} \quad (7)$$

where $(\hat{s}_{it+h} - s_{it})^+ = \hat{s}_{it+h} - s_{it} - T^{-1} \sum_{t=1}^T (\hat{s}_{it+h} - s_{it})$, $1(q_{it} \leq \gamma)^+ = 1(q_{it} \leq \gamma) - T^{-1} \sum_{t=1}^T 1(q_{it} \leq \gamma)$, $(i_{h,it} - i_{h,it}^*)^+ = (i_{h,it} - i_{h,it}^*)1(q_{it} \leq \gamma) - T^{-1} \sum_{t=1}^T (i_{h,it} - i_{h,it}^*)1(q_{it} \leq \gamma)$, $(i_{h,it} - i_{h,it}^*)^\pm = (i_{h,it} - i_{h,it}^*)1(q_{it} > \gamma) - T^{-1} \sum_{t=1}^T (i_{h,it} - i_{h,it}^*)1(q_{it} > \gamma)$, and $e_{it+h}^+ = e_{it+h} - T^{-1} \sum_{t=1}^T e_{it+h}$; note that this latter equation is simply the original threshold panel regression model (5) after removing individual-specific means.¹⁰

Next, let

¹⁰ Note that given the nature of the panel threshold model, α_1 and α_2 cannot be recovered directly. That is, because the within transformation produces $\alpha_1(1(q_{it} \leq \gamma) - T^{-1} \sum_{t=1}^T 1(q_{it} \leq \gamma)) + \alpha_2(1(q_{it} > \gamma) - T^{-1} \sum_{t=1}^T 1(q_{it} > \gamma)) = (\alpha_1 - \alpha_2)(1(q_{it} \leq \gamma) - T^{-1} \sum_{t=1}^T 1(q_{it} \leq \gamma))$, whereas we use the fact that $1(q_{it} > \gamma) = 1 - 1(q_{it} \leq \gamma)$.

$$X_i(\gamma) = \begin{bmatrix} 1(q_{i1} \leq \gamma)^+ (i_{h,i1} - i_{h,i1}^*)^+ (i_{h,i1} - i_{h,i1}^*)^\pm \\ 1(q_{i2} \leq \gamma)^+ (i_{h,i2} - i_{h,i2}^*)^+ (i_{h,i2} - i_{h,i2}^*)^\pm \\ \vdots \\ 1(q_{iT} \leq \gamma)^+ (i_{h,iT} - i_{h,iT}^*)^+ (i_{h,iT} - i_{h,iT}^*)^\pm \end{bmatrix};$$

$$Y_i = \begin{bmatrix} (\hat{s}_{i1+h} - s_{i1})^+ \\ (\hat{s}_{i2+h} - s_{i2})^+ \\ \vdots \\ (\hat{s}_{iT+h} - s_{iT})^+ \end{bmatrix}; \text{ and } e_i^+ = \begin{bmatrix} e_{i1+h}^+ \\ e_{i2+h}^+ \\ \vdots \\ e_{iT+h}^+ \end{bmatrix};$$

be the stacked mean corrected data and error terms for a given country i . Then, the stacked data and errors for the panel data model (7) are given by

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}; X(\gamma) = \begin{bmatrix} X_1(\gamma) \\ X_2(\gamma) \\ \vdots \\ X_n(\gamma) \end{bmatrix}; \text{ and } e = \begin{bmatrix} e_1^+ \\ e_2^+ \\ \vdots \\ e_n^+ \end{bmatrix}.$$

The estimation procedure involves several steps starting from a given γ , within the empirical support of the threshold variable which in our case is the uncertainty variable. The coefficients $\alpha_1 - \alpha_2$, β_1 , and β_2 can then be estimated using ordinary least squares, conditional on the given value for γ

$$\hat{\theta}(\gamma) = (X(\gamma)'X(\gamma))^{-1}X(\gamma)'Y, \quad (8)$$

where $\theta = (\alpha_1 - \alpha_2, \beta_1, \beta_2)'$, and the regression residuals are given by

$$\hat{e}(\gamma) = Y - X(\gamma)\hat{\theta}(\gamma); \quad (9)$$

finally, the sum of squared errors to be minimized is

$$S(\gamma) = \hat{e}(\gamma)'\hat{e}(\gamma). \quad (10)$$

The minimization of this sum of squared errors is carried out by using a grid search over the threshold variable space, as proposed by Hansen (2000). This involves constructing an evenly spaced grid on the empirical support of uncertainty, q_{it} , and minimizing the concentrated sum of squared errors (10). Finally, once $\hat{\gamma}$ the optimal threshold parameter, is obtained, the slope coefficient estimates are $\hat{\alpha}_1 - \hat{\alpha}_2 = \hat{\alpha}_1(\hat{\gamma}) - \hat{\alpha}_2(\hat{\gamma})$, $\hat{\beta}_1 = \hat{\beta}_1(\hat{\gamma})$, and $\hat{\beta}_2 = \hat{\beta}_2(\hat{\gamma})$. Note that other more conventional gradient algorithms are not applicable to this case as the criterion function (10) is generally not smooth.

2.2.2 Inference

When there is a threshold effect ($\alpha_1 \neq \alpha_2$) or ($\beta_1 \neq \beta_2$), then the threshold estimate $\hat{\gamma}$ is a consistent estimator for γ_0 (the true value of γ), and it has an asymptotic distribution, which is nonstandard (Hansen, 2000). Thus, the best way to produce confidence

intervals for the threshold parameter is to form the no-rejection region using the likelihood ratio statistic for the test on $\hat{\gamma}$ (Hansen, 2000). To test the null hypothesis $H_0: \gamma = \gamma_0$, the likelihood ratio test is used to reject large values of $LR(\gamma_0)$ where

$$LR(\gamma) = nT \frac{S(\gamma) - S(\hat{\gamma})}{S(\hat{\gamma})}, \quad (11)$$

where $S(\gamma)$ is defined in (10), and nT is the sample size.

The LR test converges in distribution as $n \rightarrow \infty$, for a fixed T , to a random variable ξ with distribution function $P(\xi \leq z) = (1 - \exp(-z/2))^2$. Furthermore, the distribution function ξ has the inverse

$$c(\rho) = -2\ln(1 - \sqrt{1 - \rho}), \quad (12)$$

where ρ is the significance level. The “no-rejection region” for a confidence level $1 - \rho$ is the set of values of γ such that $LR(\gamma) \leq c(\rho)$. This is found by plotting $LR(\gamma)$ against γ and drawing a flat line at $c(\rho)$.

With regard to the estimates of the slope parameters, the panel threshold regression model conditional on a given threshold parameter is just a linear regression model. Therefore, the asymptotic distribution of the estimates of the slope parameters converges to the traditional normal distribution as $n \rightarrow \infty$, either for a fixed T or $T \rightarrow \infty$.

2.2.3 Testing for threshold effects

It is critical to determine whether the threshold effects are statistically significant or not. The null hypothesis of no threshold effects in (4) can be represented by the linear constraint $H_0: \alpha_1 = \alpha_2$ and $\beta_1 = \beta_2$. However, under the null hypothesis, H_0 , the threshold γ is not identified, so classical tests have non-standard distributions. For this reason, Hansen (1999, 2000), suggests a bootstrap to simulate the asymptotic distribution of the likelihood ratio test for this model so that the p -values constructed from the bootstrap procedure are asymptotically valid.

Hence, under the null hypothesis of no threshold, the panel data model (4) is

$$\hat{s}_{it+h} - s_{it} = \mu_i + \alpha_1 + \beta_1(i_{h,it} - i_{h,it}^*) + e_{it+h}, \quad (13)$$

where the parameter β_1 can be estimated using ordinary least squares (after making the within transformation), yielding an estimate of $\hat{\beta}_1$, and residuals $\hat{e}$. Let $S_0 = \hat{e}'\hat{e}$ be the sum of squared residuals of the linear panel data model. In this case, the likelihood ratio test of H_0 is based on

$$F = nT \frac{S_0 - S(\hat{\gamma})}{S(\hat{\gamma})}; \quad (14)$$

moreover, the null hypothesis is rejected if the percentage of draws for which the simulated statistic exceeds the actual value is less than a given critical value.

2.3 Data

We collect data for a sample of advanced and emerging market countries. The country and time composition of this sample is determined by data availability. In particular, the availability of country-specific economic policy uncertainty measures is a key constraint. The need to have a balanced panel is another important restriction as the threshold model cannot yet be estimated for unbalanced panels.

Our monthly dataset comprises information for fourteen countries or regions over the period 2003:1–2021:12. The list of countries includes Australia, Canada, Chile, China, Colombia, the European Union (which is treated as a single country), Hong Kong, India, Japan, South Korea, Mexico, Russia, Singapore, and the United Kingdom. Our baseline regression requires information on the following variables: spot bilateral exchange rates, survey-based bilateral exchange rate forecasts, three-month deposit interest rates on domestic and foreign currencies, and domestic and foreign economic policy uncertainty. Foreign currency refers in this study to the U.S. dollar, which is the anchor or reference currency for the countries in the sample (see, for instance, Ilzetzki et al., 2019). The earliest economic policy uncertainty data available for some countries is 2003:1.

The spot exchange rates and the three-month domestic and foreign currency deposit interest rates were obtained from Bloomberg and Datastream, the exchange rate forecasts from Consensus Forecast, and the domestic and foreign country uncertainties from the Economic Policy Uncertainty website. Table 1 reports summary statistics for each country, averages across countries, and pooled data.

As in several UIP studies, we use the deposit interest rate in domestic currency since residents usually favor such deposits. One concern with these rates is that they can be affected by capital controls, regulations, and local taxes. However, with the advent of financial integration, these distortions have become less important. This is particularly true for the sample of countries and periods included in this study. In other studies (see, for example, Bussière et al., 2022), offshore interest rates are utilized. Unfortunately, these interest rates are only available for a limited number of countries.

As noted earlier, we focus on the *ex-ante* Fama regression model. We do so because we are using survey-based exchange rate expectations and, thus, we drop the assumption of rational expectations. Other papers that use survey-based exchange rate data include Bussière et al. (2022) and Chinn and Frankel (2019). The survey-based exchange rate expectations are obtained from the Consensus Forecast. The period t exchange rate forecast in country i is the average of the exchange rate forecasts for period t provided by each survey participant country i . The participants of the surveys of each Consensus Forecast report include several international and local economists. Some of these economists represent global firms, such as Goldman Sachs, JP Morgan, and HSBC; regional branches, such as Citigroup Japan; and other country-specific entities, such as the University of Toronto in the case of Canada.

The consensus forecast has been widely used in many empirical studies. This survey data has been found to be more precise than the random walk forecast and the forward rate forecast (Novotný & Raková, 2011) and it does not follow rules of

Table 1 Summary statistics

Variable	Expected exchange rate change				Interest rate differentials				Economic uncertainty			
	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
Australia	4.7	10.4	-26.3	37.0	2.0	1.8	-1.2	5.1	108.2	61.3	25.7	337.0
Canada	1.1	6.5	-18.8	23.1	0.2	0.8	-2.0	2.1	193.2	118.7	40.4	678.8
Chile	1.2	11.4	-45.4	27.4	1.9	1.9	-0.4	6.5	122.8	74.9	31.6	454.6
China	-1.3	5.2	-13.6	13.6	1.5	3.4	-11.3	7.3	257.3	237.1	26.1	970.8
Colombia	4.0	14.7	-46.2	52.6	3.9	1.7	0.5	8.6	109.4	48.5	41.1	376.8
European Union	2.0	7.4	-14.7	21.7	-0.5	1.2	-3.1	2.1	168.6	70.9	47.7	433.3
Hong Kong	0.4	0.8	-1.5	2.6	-0.3	0.5	-2.2	1.0	141.8	69.9	31.5	425.4
India	-0.7	6.3	-23.5	14.3	5.1	2.4	0.4	10.6	91.4	49.7	23.4	283.7
Japan	-0.4	9.9	-28.0	24.6	-1.5	1.5	-5.1	0.0	107.2	33.9	48.7	239.1
Korea	-1.9	8.2	-39.0	24.7	1.6	1.5	-0.9	4.9	146.3	69.9	37.3	538.2
Mexico	1.2	11.6	-40.4	30.3	3.6	1.1	0.2	5.9	73.5	51.1	8.5	428.7
Russia	-0.2	12.7	-49.0	69.4	3.6	2.6	-1.7	12.6	117.3	126.4	24.1	793.6
Singapore	-0.2	5.9	-14.7	20.3	-0.5	0.8	-3.4	0.6	145.1	75.3	50.4	414.8
United Kingdom	2.1	7.1	-16.9	22.3	0.4	1.3	-1.9	6.2	233.0	156.1	30.5	1141.8
Average	0.8	8.4	-27.0	27.4	1.5	1.6	-2.3	5.2	148.2	88.8	33.4	536.9
Pool	0.9	9.3	-49.4	69.4	1.5	2.6	-11.3	12.6	148.2	114.9	8.5	1141.8

Annualized values (percent). SD stands for standard deviation

thumb related to interest rate differentials or the forward exchange rate (Stavrakeva & Tang, 2015). Moreover, Batchelor (2001) documented that the Consensus Economics provide better forecasts—in terms of mean absolute error and root mean square error—than those provided by the Organization for Economic Co-operation and Development and the International Monetary Fund.

Uncertainty is proxied by the news-based index of economic policy uncertainty (EPU). This index (Baker et al., 2016), originally developed for the U.S., reflects newspaper reporting frequency of the following three items: (1) “economy” or “economic”; (2) “uncertain” or “uncertainty”; and (3) “deficit”, “Federal Reserve”, “legislation” or “White House”. The index was relatively low in the run-up to the Great Recession, with an average of 90 in the period 2003:1–2008:8, and remained relatively high during the recovery from this recession, with an average of 160 in the period 2008:9–2021:12. Country uncertainty indexes have been constructed for twenty-four countries around the world. Several of these countries, however, are members of the European Union. The EPU is available for some countries starting only in 2003:1. Largely because of this we arrive at a sample of fourteen countries over the period 2003:1–2021:12.

To assess the robustness of our main findings, we use information about the relative yield curve, capital controls, exchange rate regimes, inflation rates, and terms of trade. These variables are likely to affect the time-varying risk premium (Colacito et al., 2020; Farhi & Werning, 2014; Lloyd & Marin, 2020; Lustig et al., 2019) and, thus, the UIP in a framework with risk-averse agents. We calculate each country’s yield curve as the difference between a 10-year yield and a 6-month yield of zero-coupon sovereign bonds. The relative yield curve is the difference between a country’s yield curve and that of the base country, the U.S. Nominal yield data is mainly obtained from Bloomberg. In the cases of Colombia, Chile, and Russia, a few missing years were extrapolated to the beginning of the sample with data from their respective central banks.¹¹

The capital controls information is obtained from Fernández et al. (2016). In particular, we use the overall restrictions index which is an average of the overall inflow restrictions index and the overall outflow restrictions index. The values of the overall restrictions index are between 0 and 1, where 0 means that there are no capital inflow or outflow restrictions, and 1 means that there are many capital inflow and outflow restrictions. Given that capital inflow or outflow restrictions are slow-moving variables, we have assumed that the monthly data is the same as the yearly data.

The exchange rate regime data is obtained from Ilzetzki et al. (2019). They use a two-step procedure to determine the *de facto* degree of exchange rate flexibility. First, they determine the anchor currency for each country with the help of algorithms based on exchange rate volatility. Second, they produce a measure of exchange rate flexibility using the information on the parallel exchange market or the unified exchange rate market. Then, they classify exchange rates based on the probability of the parallel (unified) exchange rate being outside a threshold within

¹¹ The slope between the 10-year and 1-year yields and between the 10-year and 1-year sovereign bond rates (no zero-coupon) were utilized for Colombia and Chile, respectively.

a five-year window, provided that the inflation rate is not higher than 40%. We have used the fine measure of exchange rates with values between 0 and 15.

Inflation rates are calculated based on each country's consumer price index (CPI), obtained from the IMF's International Finance Statistics. For most countries, CPI data is collected with a monthly frequency, which allows us to calculate the twelve-month inflation rates.¹² Then, we use this information to construct the inflation differential of each country vis-à-vis its anchor country. Lastly, information on each country's commodity terms of trade is obtained from Gruss and Kebhaj (2019). They construct this database using the international price variation of up to forty-five commodities, which are rolling weighted using the information on each country's commodity trade data. This data is monthly and covers the period 1980:1–2021:12.

3 An empirical investigation of uncertainty and UIP

In this section, we discuss our main empirical findings on the relationship between economic policy uncertainty and the UIP. However, before doing so, we need to assess whether the key variables in the Fama regression equation are stationary or not, as this can affect the inference. We have a panel dataset where the time dimension, T , is much bigger than the cross-section dimension, N (228 months for 14 countries). Table 2 reports the unit-root tests we have performed. In the panel unit root test cases there is no evidence of a common unit root process. In the individual country unit root test cases, there is no evidence of unit roots except in one of the cases. These latter tests usually exhibit more power than the previous one as each variable is assumed to follow a unit root process under the null hypothesis. A similar finding is obtained from tests that consider cross-section dependence. On this basis, we conclude that the key variables in the Fama regression model are stationary.

In principle, economic policy uncertainty could arise from two important sources: domestic or foreign. Both types of uncertainty could potentially have different implications over the Fama regression equation. Although one could construct several proxies for uncertainty, in our baseline scenarios, we utilize domestic (Country EPU), foreign (Anchored or U.S. EPU), and weighted average uncertainty. In the latter case, the weights for each country are chosen so as to minimize the sum of squared residuals of a country-threshold Fama regression model. We utilize two weighted averages: the first one (Weighted 1 EPU) uses the weights obtained from country regressions covering the period 2003:01–2021:12 and the second one (Weighted 2 EPU) uses the weights obtained from country regressions using the longest possible span, which for some countries starts before 2003.

¹² For Australia, we convert the quarterly data to monthly data.

Table 2 Panel unit-root tests (*p*-values)

Method	Expected exchange rate change	Observed exchange rate change	Interest rate differentials	Economic uncertainty
<i>Panel unit root tests with cross-section independence</i>				
Null hypothesis: unit root (assumes individual unit root process)				
Im, Pesaran and Shin W-stat	0.0000	0.0000	0.0060	0.0000
ADF-Fisher Chi-square	0.0000	0.0000	0.0058	0.0000
PP-Fisher Chi-square	0.0000	0.0000	0.0067	0.0000
Null hypothesis: Unit root (assumes common unit root process)				
Levin, Lin and Chu <i>t</i> -stat	0.0000	0.0000	0.1275	0.0000
Breitung <i>t</i> -stat	0.0000	0.0000	0.0014	0.2493
<i>Panel unit root tests with cross-section dependence</i>				
Null hypothesis: no cointegration among all cross-sections				
Bai and Ng	0.0000	0.0000	0.0000	0.0000
Null hypothesis: unit root				
Pesaran	< 0.01	< 0.01	< 0.05	< 0.01

The tests were performed with an intercept in the specification. For the optimal lag length selection, the Akaike information criterion was used. A pooled test of the idiosyncratic errors of Bai and Ng can be seen as a panel test of no cointegration, as the null hypothesis that the error for every unit is unit-root only if no stationary combination of the data can be formed

3.1 Estimation with an imposed *ad hoc* threshold

We start by reporting the panel regression results using an arbitrarily chosen uncertainty threshold as in Ismailov and Rossi (2018). Table 3 reports the results of splitting the regression sample into two arbitrary uncertainty environments: “high uncertainty”, which corresponds to the top quartile of the uncertainty proxy; and “low uncertainty,” which includes the rest of the observations. In the discussion of the main findings of this exercise, we focus on the slope coefficient.

The first column of this Table reports the results of estimating an *ex-ante* panel Fama regression equation where uncertainty does not play any role. Consistent with the previous literature (Chinn & Frankel, 2019), the slope coefficient is positive (0.67) but significantly different from one. Thus, the null of $\beta = 1$ is rejected. Next, when we split the sample into high and low uncertainty environments, we find that in the “low uncertainty” environment, the estimates of the slope coefficients are positive but are not significantly different from one in two of the four cases considered. In contrast, in the “high uncertainty” environment, the estimates of the slope coefficient are generally positive but significantly different from one. These results are broadly in line with those reported in Ismailov and Rossi (2018).

These findings, however, raise important questions. First, is it justified to split the sample into two *ad hoc* uncertainty environments? In other words, is the threshold Fama regression model superior to the Fama regression model? Second, is the

Table 3 Panel data estimation with an imposed threshold

	Linear estimation	Threshold estimation			
		Country EPU	Anchored EPU	Weighted 1 EPU	Weighted 2 EPU
$\hat{\beta}$	0.666*** (0.113)	–	–	–	–
$\hat{\alpha}_1 - \hat{\alpha}_2$	–	– 0.006 (0.004)	– 0.003 (0.004)	– 0.008** (0.004)	– 0.007* (0.004)
<i>Low uncertainty</i>					
$\hat{\beta}_1$	–	0.841 (0.104)	0.777* (0.121)	0.835 (0.123)	0.785* (0.123)
<i>High uncertainty</i>					
$\hat{\beta}_2$	–	0.071*** (0.282)	0.165*** (0.167)	– 0.049*** (0.179)	0.005*** (0.191)
<i>Ad hoc threshold</i>	–	179.179	169.236	172.987	182.014
Observations	3192	3192	3192	3192	3192
Countries	14	14	14	14	14
Period	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12

Heteroscedasticity and autocorrelation consistent (HAC) standard errors are in parentheses, lag length is set to $T^{\frac{1}{4}}$

*, ** and *** denote statistical significance at the 10, 5 and 1% level, respectively, for the individual null hypotheses of the difference between regime intercepts equal to zero and a slope equal to 1. Weighted 1 and Weighted 2 are computed by minimizing the sum of squared residuals for each country in its own and the common samples, respectively

optimal uncertainty threshold value which splits the sample between the two uncertainty environments given by the top uncertainty quartile as in Ismailov and Rossi (2018)? This latter issue is important as Donayre (2014) shows that if the threshold is misspecified, tests for asymmetry have low power, leading to an inability to reject the null hypothesis of linearity. Third, can the presence of more than two uncertainty environments be ruled out? In the next subsection, we discuss these questions.

3.2 Test for threshold effects

Are there uncertainty threshold effects in the Fama regression equation? To address this question, we need to test for the existence of a threshold effect in the 3-month panel Fama regression equation using the F test given in Eq. (14). This step typically involves estimating Eq. (5) and computing the residual sum of squares for the different uncertainty threshold(s). With this information, we then test for the existence of uncertainty threshold effects using 1000 bootstrap replications.

Table 4 Tests for threshold effects

	Threshold estimate	Test F	Bootstrap p value	Critical values
Country uncertainty as a threshold variable	126.265	81.063	0.000	$30.360^{1/}$ $38.657^2/$ $56.577^3/$
Anchored uncertainty as a threshold variable	126.626	66.113	0.035	$45.830^{1/}$ $59.392^2/$ $81.054^3/$
Weighted 1 uncertainty as a threshold variable	144.765	70.591	0.007	$35.870^{1/}$ $43.110^2/$ $61.962^3/$
Weighted 2 uncertainty as a threshold variable	132.788	71.351	0.006	$33.496^{1/}$ $41.246^2/$ $62.612^3/$

1/, 2/ and 3/ are the critical values at 10%, 5% and 1%, respectively
 1000 bootstrap replications have been performed for each test

Table 5 Asymptotic confidence interval in threshold model

Threshold variable	Threshold estimate	90% confidence interval	99% confidence interval
Country uncertainty	126.266	[113.367; 127.821]	[111.036; 128.367]
Anchored uncertainty	126.626	[126.625; 134.402]	[120.544; 135.880]
Weighted 1 uncertainty	144.765	[119.955; 147.589]	[109.095; 153.870]
Weighted 2 uncertainty	132.788	[127.576; 133.841]	[126.641; 134.643]

The results of the test for uncertainty threshold effects are shown in Table 4. The null hypothesis of no threshold effect against a single threshold can be rejected in all cases at the 5% significance level, and in all but one case at the 1% significance level. For instance, the F test statistic for the existence of a single threshold when using the domestic (foreign) uncertainty as a threshold variable is 81.1 (66.1), with a bootstrap p -value of 0.000 (0.035). Together these results indicate that the tests for an uncertainty threshold are highly significant, regardless of the uncertainty proxy that has been used. Therefore, there is strong evidence that uncertainty affects the Fama equation by splitting the regression sample into two regimes.¹³ In addition, we perform tests for the existence of two or more threshold effects, which implies that the sample should be split into three or more uncertainty regimes. These additional tests, which are not reported here, suggest that there are no additional thresholds beyond those tabulated in Table 4.

3.3 Confidence interval of the threshold estimate

Next, we construct a confidence interval for the estimated uncertainty threshold. In particular, the point estimates of the threshold parameter and their asymptotic 90 and 99% confidence intervals are reported in Table 5 for the four uncertainty proxies. The two regimes generated by the threshold estimate are denoted low and high uncertainty regimes, respectively. The asymptotic confidence intervals for the threshold parameter are tight, which indicates high precision in the estimation. Perhaps more importantly, note that the threshold estimates and their corresponding confidence intervals are much smaller than those obtained under the *ad hoc* rule (Table 3), suggesting the unfitness of the latter. This shows the superiority of the framework that we are using.

Additional information about the threshold estimates and their confidence intervals is provided in Fig. 1. We obtain the two panels by plotting the concentrated likelihood ratio function $LR(\gamma)$ of the threshold parameter when using the domestic and foreign uncertainty proxies, respectively. The function is minimized at zero when the estimated thresholds are $\hat{\gamma} = 126.3$ and $\hat{\gamma} = 126.6$. We obtain similar results when using the weighted uncertainty measures. The estimation precision is

¹³ Note, we are using the term uncertainty *regimes* as opposed to uncertainty *environments* used in Sect. 3.1, to imply that, in this case, the uncertainty thresholds have been selected optimally.

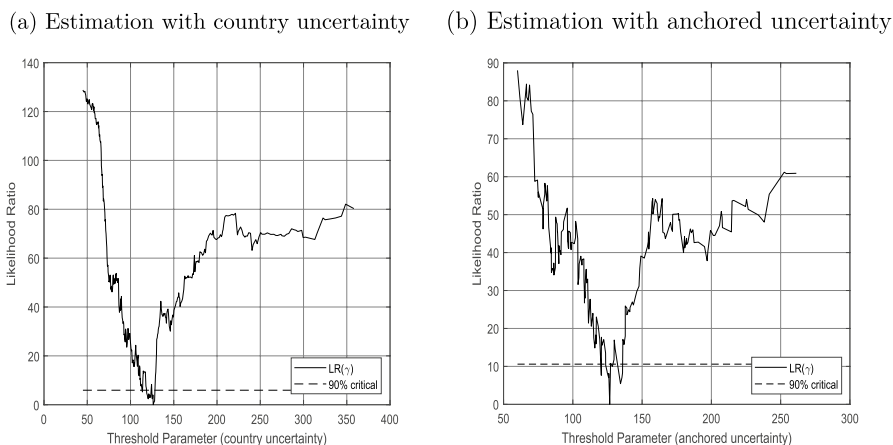


Fig. 1 Confidence interval construction for threshold

high because the 90% confidence interval, the set of values below the dotted line in Fig. 1, is rather narrow across the threshold parameter space.

3.4 Slope estimation results

In the panel threshold Fama regression framework, the slope coefficients in the two uncertainty regimes should be different. We find that the slope coefficients are much different across regimes as reported in Table 6. For instance, the slope parameter estimate in the low uncertainty regime is slightly greater than one. Moreover, this estimate is not significantly different from one, at the 1% level, regardless of the uncertainty proxy. In contrast, the estimate of the slope parameter in the high uncertainty regime is generally positive but it is significantly different from one in all cases. This finding confirms our *a priori* belief that the deviations from UIP tend to be more severe in the more uncertain regimes.

The evidence suggests that the panel threshold Fama regression model should be preferred over the panel linear Fama regression model. The null hypothesis of a linear Fama model is rejected, in all cases, in favor of a threshold regression Fama model with two regimes, whereby the slope estimates in each regime are statistically different from each other. In addition, the estimate of the difference between intercepts in the two uncertainty regimes is significantly different from zero in three out of the four uncertainty proxies, providing complementary evidence in favor of the panel threshold regression model.

It is also worth noting that the point estimates of the slope coefficient in the threshold regression models are slightly higher and closer to one than those observed using the *ad hoc* threshold model. Moreover, the slope parameter estimates in the threshold regression framework for the low uncertainty regime are not significantly different from one, independent of the uncertainty proxy utilized, while half of the slope parameter estimates in the low uncertainty environment in the *ad hoc* threshold model are significantly different from one.

Table 6 Panel data estimation with an estimated threshold

	Linear estimation	Threshold estimation			
		Country EPU	Anchored EPU	Weighted 1 EPU	Weighted 2 EPU
$\hat{\beta}$	0.666** (0.113)	—	—	—	—
$\hat{\alpha}_1 - \hat{\alpha}_2$	—	− 0.007* (0.004)	− 0.003 (0.003)	− 0.008*** (0.003)	− 0.009*** (0.003)
<i>Low uncertainty</i>					
$\hat{\beta}_1$	—	1.184 (0.116)	1.008 (0.136)	1.027 (0.129)	1.085 (0.136)
<i>High uncertainty</i>					
$\hat{\beta}_2$	—	0.060*** (0.170)	0.088*** (0.133)	− 0.062*** (0.152)	0.036*** (0.137)
Threshold estimate	—	126.266	126.626	144.765	132.788
90% confidence interval	—	[113.4, 127.8]	[126.6, 134.4]	[120.0, 147.6]	[127.6, 133.8]
Test for threshold effects	—	0.000	0.035	0.007	0.006
Observations	3192	3192	3192	3192	3192
Countries	14	14	14	14	14
Period	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12

Heteroscedasticity and autocorrelation consistent (HAC) standard errors are in parentheses, lag length is set to $T^{\frac{1}{4}}$. The test for threshold effects shows the probability value for the null hypothesis of $\hat{\alpha}_1 = \hat{\alpha}_2$ and $\hat{\beta}_1 = \hat{\beta}_2$; we used 1000 bootstrap replications for the test

*, **, and *** denote statistical significance at the 10, 5 and 1% level, respectively, for the individual null hypotheses of the difference between regime intercepts equal to zero and a slope equal to 1

Table 7 Percentage of observations in each regime by country

Variable	Country uncertainty		Anchored uncertainty		Weighted 1 uncertainty		Weighted 2 uncertainty	
	Low	High	Low	High	Low	High	Low	High
Australia	71.1	28.9	54.4	45.6	68.9	31.1	58.8	41.2
Canada	33.3	66.7	54.4	45.6	58.3	41.7	40.4	59.6
Chile	67.7	33.3	54.4	45.6	75.0	25.0	70.2	29.8
China	41.2	58.8	54.4	45.6	47.4	52.6	43.9	56.1
Colombia	68.9	31.1	54.4	45.6	74.1	25.9	66.7	33.3
European Union	30.3	69.7	54.4	45.6	48.7	51.3	31.6	68.4
Hong Kong	47.8	52.2	54.4	45.6	61.0	39.0	53.5	46.5
India	79.4	20.6	54.4	45.6	68.0	32.0	59.2	40.8
Japan	76.3	23.7	54.4	45.6	70.6	29.4	60.1	39.9
South Korea	44.7	55.3	54.4	45.6	60.5	39.5	49.1	50.9
Mexico	92.1	7.9	54.4	45.6	94.3	5.7	87.7	12.3
Russia	46.9	53.1	54.4	45.6	56.6	43.4	51.3	48.7
Singapore	49.6	50.4	54.4	45.6	61.0	39.0	53.5	46.5
United Kingdom	28.1	71.9	54.4	45.6	33.8	66.2	30.3	69.7
Full sample	55.5	44.5	54.4	45.6	62.7	37.3	54.0	46.0

How sensitive are our main empirical findings to including additional controls, use of *ex-post* exchange rates, changes in term maturity, and estimation methods? Appendix A reports the results of a series of sensitivity exercises where (1) additional regressors are included in the uncertainty threshold Fama regression: uncertainty itself (as a proxy for risk premia), yield curve slope differential, capital controls, inflation differentials, terms of trade, and time-fixed effects (controlling for omitted time-varying variables not available, e.g. transaction costs, CIP deviations); (2) *ex-post* UIP estimation to assess the importance of survey-based exchange rates; (3) change in the maturity of the asset to one-year instead of 3-month; and (4) change in the estimation method to maximum likelihood instead of ordinary least squares. In all these sensitivity exercises, the results align well with the baseline results, where the UIP condition holds in the low-uncertainty regime but fails in the high-uncertainty one.

3.5 Importance of high and low uncertainty regimes

These findings beg the following questions: what fraction of the observations belong to each uncertainty regime? Which countries are more susceptible to experiencing a high uncertainty regime? Table 7 provides information that addresses these questions for the different proxies of uncertainty. First, for the domestic and weighted 1 average uncertainty proxies, the model establishes that between 44.5 and 37.3% of the observations in the sample belong to the

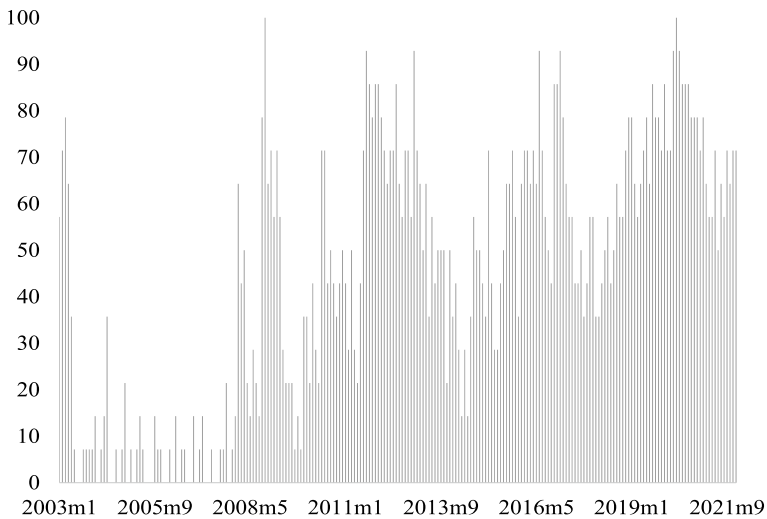


Fig. 2 Percentage of countries in a high uncertainty regime over time

high uncertainty regime. This fraction is much higher than the 25% assumed in Ismailov and Rossi (2018). The foreign uncertainty proxy, similarly, suggests that 45.6% of the sample corresponds to the high uncertainty regime.

Second, in terms of the individual countries, we note that more than 55% of the observations for Canada, China, the European Union, South Korea, and the United Kingdom are in the high country uncertainty regime when the country uncertainty is used for classification. In contrast, more than 60% of the observations for Australia, Chile, Colombia, India, Japan, and Mexico are in the low country uncertainty regime.

What is the time pattern of these uncertainty regimes? Fig. 2 shows the evolution over time of the fraction of countries in the high uncertainty regime. We focus on the case where we use the domestic uncertainty proxy. The chart suggests that the low uncertainty regime is mostly concentrated in the period of the run-up to the Great Recession, except for a short bout of uncertainty in 2003, which is consistent with the literature on the Great Moderation. The high uncertainty regime is, in contrast, concentrated in the period around and following the Great Recession. This is certainly the case in Canada, the European Union, and the United Kingdom. The post-Great Recession period, however, does not show a uniform pattern as months of high uncertainty are followed by months of low uncertainty. Bussière et al. (2022) also report a break in their Fama regression results, starting with the great financial crisis. However, they do not link this break to uncertainty since they use a linear model.

4 Conclusion

In this paper, we study whether uncertainty can help explain the uncovered interest parity puzzle. Because of the subtle effects of uncertainty on financial markets, we postulate that uncertainty and the UIP condition could be usefully modeled using a panel threshold regression model, with uncertainty as a threshold variable. Using survey-based exchange rate expectations, we find that there is a statistically significant uncertainty threshold that splits the sample into two regimes: a low-uncertainty regime and a high-uncertainty regime. Most importantly, our analysis finds that the UIP condition holds in the low-uncertainty regime but does not in the high-uncertainty one. This suggests that both the linearity of the Fama regression model and the omission of uncertainty from this model might be at the core of the negative empirical results widely reported in the literature.

Why does UIP hold in the low-uncertainty regime but not in the high-uncertainty one? As we have discussed, there is research that highlights the important asymmetric effects of uncertainty on financial markets. During high-uncertainty periods one observes a tightening in credit conditions, higher transaction costs, uncertain arbitrage profit opportunities, and collapses in asset trades. Higher excess returns in currency carry trade operations are also possible, and the UIP condition is not likely observed. In contrast, none of these effects are present during low-uncertainty periods, and the UIP condition is likely to hold. While recent analysis has examined the differences between the uncertainty and risk aversion channels, further work is needed to grasp the complexities of uncertainty fully. In particular, the welfare implications of the effects that uncertainty has on the financial markets are not yet well understood (Rehse et al., 2019).

Various additional extensions are desirable, e.g., different specifications, when the omitted variables vary across the different uncertainty regimes. However, this requires progress both on the theoretical front from which one can derive the hypothesis to be tested and on the construction of suitable databases to conduct the empirical exercises.

Appendix

In this appendix, we examine the robustness of our main findings to the inclusion of additional controls, *ex-post* UIP estimation, and changes in the maturity of the asset and the estimation method. As noted earlier, some argue that the failure of UIP might be due to the omission of certain regressors that could be related to the risk premium. In this section, we examine seven of these variables: uncertainty itself, bond risk premia, capital controls, exchange rate flexibility, inflation differentials, terms of trade, and global common factors.

Including additional control variables

In a world with risk-averse investors, we must amend the uncertainty threshold panel Fama regression model to include a risk premium term. This accounts for the fact

that the covariance between the stochastic discount factor and the change in the nominal exchange rate differs from zero in the Corsetti and Marin (2020) framework and thus should be added to the cash flow from the arbitrage position in two-currency investments.

First, when uncertainty and the risk premium are positively correlated, an increase in the former should lead to UIP deviations due to the risk premium. To evaluate this possibility, we include uncertainty in the Fama regression equation both as a threshold variable and as an additional regressor.¹⁴ Table 8 reports the main results of this exercise. The coefficient of the additional uncertainty term is generally positive and significantly different from zero. However, the estimated parameter values are numerically unimportant in the relevant cases. These findings are broadly consistent with those reported by Albagli et al. (2023), namely that global uncertainty shocks appreciate the dollar but reduce interest rate differentials, exacerbating the widely documented UIP violation. In contrast, the slope parameter estimates of the Fama regression are closer to one than our benchmark estimates reported in Table 6. This suggests that uncertainty affects the Fama regression model mainly through its effect as a threshold variable and the threshold regression model is not the result of a misspecified linear regression model, as some might argue.

More recently Lustig et al. (2019) have suggested that the currency risk premia and the maturity of bonds are inversely related. A related study by Lloyd and Marin (2020) reports a tent-shaped relationship between the currency risk premia and the relative yield curve (for the 3- and 5-year horizons). Is the uncertainty threshold UIP a result of omitting the relative yield curve from the regression model? We have examined this possibility by including a country's yield curve relative to the U.S. as an additional regressor in the panel threshold UIP model.¹⁵ Here each country's yield curve is calculated as the difference between the 10-year yield and 6-month yield, as explained in the data section.

Table 9 shows the results of this exercise. The main takeaway from this Table is that including the relative yield curve as an additional regressor does not significantly alter the threshold UIP coefficient estimates. In contrast, the results for the estimated coefficient of the relative yield curve are mixed—it is virtually zero and insignificant in the linear model, and it is negative and statistically significant in the threshold model (except when using the anchored uncertainty proxy in which case it is not significant). This finding is broadly consistent with the one reported by Lloyd and Marin (2020), who also report a negative but insignificantly relative yield curve coefficient for a 10-year horizon.

Next, we examine the effects of capital controls on the expected change in the nominal exchange rate. Capital controls have been used by countries to mitigate the exchange rate effects of a Sudden Stop or a capital inflow surge. Farhi and Werning

¹⁴ Bussière et al. (2022) include the VIX as an additional regressor in the Fama regression equation. The risk premium affects not only the nominal exchange rate but also investment, inflation, and fiscal variables through debt service (André et al., 2023).

¹⁵ That is, the UIP regression is augmented with a measure of the relative yield curve slope $Y_{i,t} - Y_{i,t}^*$, where $Y_{i,t}$ is the slope of the base-country yield curve and $Y_{i,t}^*$ is the slope of the foreign anchor country yield curve, where the anchor country is the U.S.

Table 8 Panel data estimation with uncertainty as a regressor

	Linear estimation	Threshold estimation			
		Country EPU	Anchored EPU	Weighted 1 EPU	Weighted 2 EPU
Uncertainty/1000	0.007 (0.017)	0.037* (0.021)	0.016 (0.041)	0.060*** (0.019)	0.049*** (0.017)
$\hat{\beta}$	0.663*** (0.114)	–	–	–	–
$\hat{\alpha}_1 - \hat{\alpha}_2$	–	– 0.002 (0.005)	– 0.001 (0.005)	0.002 (0.004)	0.000 (0.004)
Low uncertainty					
$\hat{\beta}_1$	–	1.179 (0.116)	1.006 (0.136)	1.021 (0.131)	1.093 (0.137)
High uncertainty					
$\hat{\beta}_2$	–	0.074*** (0.170)	0.092*** (0.133)	– 0.001*** (0.146)	0.115*** (0.136)
Threshold estimate	–	126.266	126.626	142.516	127.940
90% confidence interval	–	[118.8, 127.8]	[126.6, 134.4]	[116.9, 147.6]	[127.6, 133.8]
Test for threshold effects	–	0.000	0.084	0.007	0.003
Observations	3192	3192	3192	3192	3192
Countries	14	14	14	14	14
Period	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12

Heteroscedasticity and autocorrelation consistent (HAC) standard errors are in parentheses, lag length is set to $T^{\frac{1}{4}}$. The test for threshold effects shows the probability value for the null hypothesis of $\hat{\alpha}_1 = \hat{\alpha}_2$ and $\hat{\beta}_1 = \hat{\beta}_2$; we used 1000 bootstrap replications for the test

*, ** and *** denote statistical significance at the 10, 5 and 1% level, respectively, for the individual null hypotheses of the difference between regime intercepts equal to zero and a slope equal to 1. The linear estimation includes the own country uncertainty; the other uncertainty measures give similar results. For the additional control, the null is that its coefficient estimate is equal to zero

Table 9 Panel data estimation with relative yield curve as a regressor

	Linear estimation	Threshold estimation			
		Country EPU	Anchored EPU	Weighted 1 EPU	Weighted 2 EPU
Yield curve slope differential	0.000 (0.002)	− 0.007* (0.004)	− 0.002 (0.003)	− 0.008*** (0.003)	− 0.009*** (0.003)
$\hat{\beta}$	0.663** (0.138)	−	−	−	−
$\hat{\alpha}_1 - \hat{\alpha}_2$	−	0.000 (0.002)	0.000 (0.002)	0.001 (0.002)	0.001 (0.002)
Low uncertainty					
$\hat{\beta}_1$	−	1.195 (0.142)	1.024 (0.157)	1.074 (0.157)	1.136 (0.163)
High uncertainty					
$\hat{\beta}_2$	−	0.070*** (0.183)	0.104*** (0.146)	− 0.020*** (0.164)	0.080*** (0.151)
Threshold estimate	−	126.266	126.626	144.765	132.788
90% confidence interval	−	[112.9, 127.8]	[126.6, 134.4]	[110.1, 147.6]	[127.6, 133.8]
Test for threshold effects	−	0.003	0.085	0.023	0.015
Observations	3192	3192	3192	3192	3192
Countries	14	14	14	14	14
Period	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12

Heteroscedasticity and autocorrelation consistent (HAC) standard errors are in parentheses, lag length is set to $T^{1/4}$. The test for threshold effects shows the probability value for the null hypothesis of $\hat{\alpha}_1 = \hat{\alpha}_2$ and $\hat{\beta}_1 = \hat{\beta}_2$; we used 1000 bootstrap replications for the test

*, ** and *** denote statistical significance at the 10, 5 and 1% level, respectively, for the individual null hypotheses of the difference between regime intercepts equal to zero and a slope equal to 1. The linear estimation includes the own country uncertainty; the other uncertainty measures give similar results. For the additional control, the null is that its coefficient estimate is equal to zero

Table 10 Panel data estimation with capital control indicator as a regressor

	Linear estimation	Threshold estimation			
		Country EPU	Anchored EPU	Weighted 1 EPU	Weighted 2 EPU
Capital control	− 0.078*** (0.028)	− 0.087*** (0.028)	− 0.079*** (0.029)	− 0.090*** (0.029)	− 0.080*** (0.029)
$\hat{\beta}$	0.601*** (0.106)	—	—	—	—
$\hat{\alpha}_1 - \hat{\alpha}_2$	—	− 0.007* (0.004)	− 0.002 (0.003)	− 0.008*** (0.003)	− 0.009*** (0.003)
<i>Low uncertainty</i>					
$\hat{\beta}_1$	—	1.119 (0.117)	0.943 (0.134)	0.960 (0.127)	1.021 (0.134)
<i>High uncertainty</i>					
$\hat{\beta}_2$	—	− 0.019*** (0.170)	0.023*** (0.136)	− 0.148*** (0.156)	− 0.032*** (0.140)
Threshold estimate	—	126.266	126.626	144.765	132.788
90% confidence interval	—	[111.7, 127.8]	[126.6, 134.4]	[109.7, 147.6]	[127.6, 133.8]
Test for threshold effects	—	0.003	0.092	0.014	0.016
Observations	3192	3192	3192	3192	3192
Countries	14	14	14	14	14
Period	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12

Heteroscedasticity and autocorrelation consistent (HAC) standard errors are in parentheses, lag length is set to $T^{\frac{1}{4}}$. The test for threshold effects shows the probability value for the null hypothesis of $\hat{\alpha}_1 = \hat{\alpha}_2$ and $\hat{\beta}_1 = \hat{\beta}_2$; we used 1000 bootstrap replications for the test

*, ** and *** denote statistical significance at the 10, 5 and 1% level, respectively, for the individual null hypotheses of the difference between regime intercepts equal to zero and a slope equal to 1. For the additional control, the null is that its coefficient estimate is equal to zero

(2014) show that this policy makes sense even in economies with flexible exchange rate regimes, as optimal capital controls take the form of temporary taxes when there are outflows and subsidies when there are inflows thus helping to mitigate the effect of these flows on the nominal exchange rate. When we include the proxy of international capital control restrictions developed by Fernández et al. (2016) as an additional regressor, we find evidence that these controls tend to appreciate the domestic currency. This effect is significantly different from zero (Table 10). Despite this result, the importance of uncertainty as a threshold variable remains virtually unchanged from our baseline regression.

How sensitive is the expected change in the nominal exchange rate to a country's nominal exchange rate regime? A large literature suggests that countries with flexible exchange rate regimes should have more depreciated exchange rates than countries with fixed exchange rates. To explore this possibility, we include the *de facto* exchange rate regime proxy, constructed by Ilzetzki et al. (2019), as an additional regressor in the Fama

Table 11 Panel data estimation with exchange rate flexibility as a regressor

	Linear estimation	Threshold estimation			
		Country EPU	Anchored EPU	Weighted 1 EPU	Weighted 2 EPU
Exchange rate flexibility	- 0.002** (0.001)	- 0.001 (0.001)	- 0.001 (0.001)	- 0.001 (0.001)	- 0.001 (0.001)
$\hat{\beta}$	0.661*** (0.110)	—	—	—	—
$\hat{\alpha}_1 - \hat{\alpha}_2$	—	- 0.007* (0.004)	- 0.002 (0.003)	- 0.008*** (0.003)	- 0.009*** (0.003)
<i>Low uncertainty</i>					
$\hat{\beta}_1$	—	1.193* (0.115)	1.016 (0.132)	1.036 (0.126)	1.092 (0.133)
<i>High uncertainty</i>					
$\hat{\beta}_2$	—	0.082*** (0.165)	0.107*** (0.134)	- 0.041*** (0.153)	- 0.054*** (0.137)
Threshold estimate	—	126.266	126.626	144.765	114.676
90% confidence interval	—	[112.9, 127.8]	[126.6, 134.4]	[110.1, 147.6]	[113.0, 129.9]
Test for threshold effects	—	0.004	0.100	0.018	0.012
Observations	3192	3192	3192	3192	3192
Countries	14	14	14	14	14
Period	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12

Heteroscedasticity and autocorrelation consistent (HAC) standard errors are in parentheses, lag length is set to $T^{\frac{1}{4}}$. The test for threshold effects shows the probability value for the null hypothesis of $\hat{\alpha}_1 = \hat{\alpha}_2$ and $\hat{\beta}_1 = \hat{\beta}_2$; we used 1000 bootstrap replications for the test

*, ** and *** denote statistical significance at the 10, 5 and 1% level, respectively, for the individual null hypotheses of the difference between regime intercepts equal to zero and a slope equal to 1. For the additional control, the null is that its coefficient estimate is equal to zero

regression equation. Table 11 reports that there is no statistically significant association between the exchange rate regime and the expected change in the nominal exchange rate, in all regression specifications but the linear panel case. This suggests that countries with more flexible exchange rate regimes do not have more depreciated exchange rates. In this exercise, uncertainty generally remains a significant threshold variable.¹⁶

Some authors argue that inflation could have an effect on the expected change in the nominal exchange rate. In particular, a country with a higher inflation rate is expected to have a more depreciated exchange rate. To examine this possibility, we include the inflation differential as another regressor in the Fama regression equation. The inflation differential is calculated as the difference between the domestic

¹⁶ This is so since the slope coefficient is now significantly different from one in the country EPU threshold regression.

Table 12 Panel data estimation with inflation differential as a regressor

	Linear estimation	Threshold estimation			
		Country EPU	Anchored EPU	Weighted 1 EPU	Weighted 2 EPU
Inflation differentials	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)
$\hat{\beta}$	0.600*** (0.101)	—	—	—	—
$\hat{\alpha}_1 - \hat{\alpha}_2$	—	− 0.007* (0.004)	− 0.003 (0.003)	− 0.008*** (0.003)	− 0.008*** (0.003)
<i>Low uncertainty</i>					
$\hat{\beta}_1$	—	1.118 (0.113)	0.950 (0.123)	0.966 (0.116)	1.029 (0.123)
<i>High uncertainty</i>					
$\hat{\beta}_2$	—	0.001*** (0.158)	0.037*** (0.131)	− 0.109*** (0.153)	0.005*** (0.139)
Threshold estimate	—	126.266	126.626	144.765	130.959
90% confidence interval	—	[111.7, 127.8]	[126.6, 134.4]	[109.1, 147.6]	[127.6, 133.8]
Test for threshold effects	—	0.002	0.077	0.018	0.016
Observations	3192	3192	3192	3192	3192
Countries	14	14	14	14	14
Period	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12

Heteroscedasticity and autocorrelation consistent (HAC) standard errors are in parentheses, lag length is set to $T^{\frac{1}{4}}$. The test for threshold effects shows the probability value for the null hypothesis of $\hat{\alpha}_1 = \hat{\alpha}_2$ and $\hat{\beta}_1 = \hat{\beta}_2$; we used 1000 bootstrap replications for the test

*, ** and *** denote statistical significance at the 10, 5 and 1% level, respectively, for the individual null hypotheses of the difference between regime intercepts equal to zero and a slope equal to 1. For the additional control, the null is that its coefficient estimate is equal to zero

and foreign inflation. We find that the coefficient of the inflation differential is positive and significantly different from zero (Table 12) as predicted by theory. In addition, uncertainty remains a robust threshold variable.

The commodity terms of trade is another variable that has been associated with the evolution of the expected change in nominal exchange rates (see Arroyo Marioli et al., 2022). According to this literature, favorable commodity prices tend to be associated with more appreciated exchange rates. To examine this possibility, we include the annual percentage change of commodity terms of trade for each country (Gruss & Kebhaj, 2019) as additional regressor. As predicted by theory there is a negative and statistically significant association between the change of commodity terms of trade and the expected change in the nominal exchange rate (Table 13). Perhaps more importantly, the status of uncertainty as a threshold variable does not change.

Table 13 Panel data estimation with terms of trade as a regressor

	Linear estimation	Threshold estimation			
		Country EPU	Anchored EPU	Weighted 1 EPU	Weighted 2 EPU
Terms of trade	− 0.002** (0.001)	− 0.002** (0.001)	− 0.002** (0.001)	− 0.002** (0.001)	− 0.002* (0.001)
$\hat{\beta}$	0.619*** (0.109)	—	—	—	—
$\hat{\alpha}_1 - \hat{\alpha}_2$	—	− 0.006 (0.004)	− 0.003 (0.003)	− 0.007* (0.003)	− 0.008** (0.003)
<i>Low uncertainty</i>					
$\hat{\beta}_1$	—	1.136 (0.114)	0.962 (0.130)	0.980 (0.124)	1.041 (0.130)
<i>High uncertainty</i>					
$\hat{\beta}_2$	—	0.022*** (0.165)	0.046*** (0.132)	− 0.104*** (0.152)	0.011*** (0.181)
Threshold estimate	—	126.266	126.626	144.765	130.960
90% confidence interval	—	[111.7, 127.8]	[126.6, 134.4]	[109.7, 147.6]	[127.6, 133.8]
Test for threshold effects	—	0.010	0.084	0.022	0.017
Observations	3192	3192	3192	3192	3192
Countries	14	14	14	14	14
Period	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12

Heteroscedasticity and autocorrelation consistent (HAC) standard errors are in parentheses, lag length is set to $T^{\frac{1}{4}}$. The test for threshold effects shows the probability value for the null hypothesis of $\hat{\alpha}_1 = \hat{\alpha}_2$ and $\hat{\beta}_1 = \hat{\beta}_2$; we used 1000 bootstrap replications for the test

*, ** and *** denote statistical significance at the 10, 5 and 1% level, respectively, for the individual null hypotheses of the difference between regime intercepts equal to zero and a slope equal to 1. For the additional control, the null is that its coefficient estimate is equal to zero

How would our main findings change if we simultaneously included all the additional controls in the threshold Fama regression equation? As reported in Table 14, we find that uncertainty remains a robust threshold variable (as the slope coefficient is not significantly different from one in the low uncertainty regime, but it is significantly different from one in the high uncertainty regime) and that the intercepts are not significantly different across uncertainty regimes. We also find some differences with the results we have just discussed. The most notable changes are the sign flip in the inflation differentials, the statistical significance of the exchange rate regime, and the statistical insignificance of the change in the commodity terms of trade. However, none of these controls are central to our analysis. Lastly, the mixed results obtained for uncertainty as an additional regressor and the statistical insignificance of the relative yield curve suggest that the omission of the exchange rate risk premia is not driving our main results.

Table 14 Panel data estimation with controls as regressors

	Linear estimation	Threshold estimation			
		Country EPU	Anchored EPU	Weighted 1 EPU	Weighted 2 EPU
Uncertainty	− 0.001 (0.017)	0.032 (0.021)	− 0.012 (0.041)	0.050*** (0.018)	0.042** (0.017)
Yield curve slope diff	− 0.001 (0.002)	− 0.001 (0.002)	− 0.001 (0.002)	0.000 (0.002)	0.000 (0.002)
Capital controls	− 0.069** (0.030)	− 0.077*** (0.029)	− 0.065** (0.029)	− 0.075** (0.030)	− 0.066** (0.029)
Exchange rate flexibility	− 0.003*** (0.001)	− 0.003*** (0.001)	− 0.002** (0.001)	− 0.002** (0.001)	− 0.002** (0.001)
Inflation dif- ferentials	− 0.003*** (0.001)	− 0.002** (0.001)	− 0.003*** (0.001)	− 0.003*** (0.001)	− 0.003*** (0.001)
Terms of trade	− 0.001 (0.001)	− 0.001 (0.001)	− 0.001 (0.001)	− 0.001 (0.001)	− 0.001 (0.001)
$\hat{\beta}$	0.611*** (0.132)	—	—	—	—
$\hat{\alpha}_1 - \hat{\alpha}_2$	—	− 0.001 (0.005)	− 0.003 (0.005)	0.002 (0.004)	0.000 (0.004)
<i>Lowuncertainty</i>					
$\hat{\beta}_1$	—	1.086 (0.146)	0.957 (0.150)	0.987 (0.152)	1.055 (0.160)
<i>High uncertainty</i>					
$\hat{\beta}_2$	—	0.021*** (0.168)	0.063*** (0.152)	− 0.034*** (0.161)	0.098** (0.148)
Threshold estimate	—	126.266	126.626	142.516	127.940
90% confidence interval	—	[113.4, 127.8]	[84.6, 134.4]	[120.0, 147.6]	[127.6, 133.8]
Test for thresh- old effects	—	0.001	0.091	0.007	0.006
Observations	3192	3192	3192	3192	3192
Countries	14	14	14	14	14
Period	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12

Heteroscedasticity and autocorrelation consistent (HAC) standard errors are in parentheses, lag length is set to $T^{\frac{1}{4}}$. The test for threshold effects shows the probability value for the null hypothesis of $\hat{\alpha}_1 = \hat{\alpha}_2$ and $\hat{\beta}_1 = \hat{\beta}_2$; we used 1000 bootstrap replications for the test

*, ** and *** denote statistical significance at the 10, 5 and 1% level, respectively, for the individual null hypotheses of the difference between regime intercepts equal to zero and a slope equal to 1. The linear estimation includes the own country uncertainty; the other uncertainty measures give similar results. For the additional controls, the null is that each coefficient estimate is equal to zero

There are other global factors that could have an effect on the Fama threshold regression model. For instance, the Great Recession, the evolution of international oil prices, global financial conditions, etc. While controlling for such variables is desirable, there are no good measures available. A proxy commonly used to capture these global factors is the use of a time-fixed effects variable, which is common to all countries in the sample. In Table 15, we report the results of including such a

Table 15 Panel data estimation with time fixed effects

	Linear estimation	Threshold estimation			
		Country EPU	Anchored EPU	Weighted 1 EPU	Weighted 2 EPU
$\hat{\beta}$	0.726** (0.131)	—	—	—	—
$\hat{\alpha}_1 - \hat{\alpha}_2$	—	− 0.018*** (0.004)	− 0.003 (0.004)	− 0.019*** (0.005)	− 0.021*** (0.005)
<i>Low uncertainty</i>					
$\hat{\beta}_1$	—	1.344* (0.134)	1.073 (0.148)	1.405*** (0.135)	1.269* (0.159)
<i>Low uncertainty</i>					
$\hat{\beta}_2$	—	0.137*** (0.179)	0.081*** (0.145)	0.242*** (0.163)	0.157*** (0.145)
Time dummies	✓	✓	✓	✓	✓
Threshold estimate	—	126.504	126.625	109.658	127.940
90% confidence interval	—	[120.5, 126.8]	[84.5, 126.6]	[109.1, 111.1]	[100.9, 133.8]
Test for threshold effects	—	0.000	0.003	0.002	0.001
Observations	3192	3192	3192	3192	3192
Countries	14	14	14	14	14
Period	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12

Heteroscedasticity and autocorrelation consistent (HAC) standard errors are in parentheses, lag length is set to $T^{\frac{1}{4}}$. The test for threshold effects shows the probability value for the null hypothesis of $\hat{\alpha}_1 = \hat{\alpha}_2$ and $\hat{\beta}_1 = \hat{\beta}_2$; we used 1000 bootstrap replications for the test

*, ** and *** denote statistical significance at the 10, 5 and 1% level, respectively, for the individual null hypotheses of the difference between regime intercepts equal to zero and a slope equal to 1

variable. While there is still evidence of a significant threshold effect, only the estimate of the slope parameter in the low uncertainty regime with the country EPU proxy is not significantly different from one. In addition, the intercept estimates are not statistically different across uncertainty regimes.

Ex-post UIP estimation

As noted before, the expected returns in domestic and foreign assets must equalize under rational expectations and risk neutrality. However, most empirical studies have reported that this UIP condition does not hold. In this study, we have found that when modeling this arbitrage condition as an *ex-ante* panel uncertainty threshold regression, the UIP holds during low-uncertainty periods but fails in high-uncertainty ones. An important question is whether our findings depend on using

the *ex-ante* exchange rate survey data instead of the often utilized *ex-post* exchange rates.

To examine this possibility, we estimate the *ex-post* panel uncertainty threshold Fama regression model. The UIP under rational expectations is the hypothesis that there is no foreign exchange risk premium, or that the expected excess return on foreign bonds equals zero (Engel et al., 2022). Thus, the *ex-post* Fama panel regression equation is given by

$$s_{it+h} - s_{it} = \mu_i + \alpha + \beta(i_{h,it} - i_{h,it}^*) + e_{it+h}, \quad (15)$$

where s_{it+h} is the realized (in logs) exchange rate in period $t + h$ for country i .

Table 16 reports the results of this exercise. First, the *ex-post* panel uncertainty threshold Fama regression estimation results are similar to those of the *ex-ante* case. In three of the four uncertainty indexes, the UIP condition holds in low-uncertainty periods but fails otherwise. However, when using the country EPU uncertainty index, the uncertainty threshold estimate is much higher, yet there is no evidence of a threshold effect. Second, the *ex-post* regression results show generally higher standard errors than those observed in the *ex-ante* case. This finding is unsurprising as the former is expected to include larger forecasting errors. In sum, this exercise suggests that the observed threshold effects in the UIP condition are not the result of using survey-based exchange expectations in the regression analysis, as similar findings are obtained when using *ex-post* exchange rates.

Estimation with one-year forecast horizon

What happens with our main findings if the threshold Fama regression model is estimated using 12-month horizon expected exchange rates and 12-month deposit rates instead of the three-month horizon variables used in our baseline?

To address this question, we utilize 12-month horizon survey-based exchange rate expectations obtained from the Consensus Forecast and the twelve-month interest rates on domestic and foreign exchange deposits obtained from the sources discussed earlier.¹⁷ Table 17 reports the main results of this exercise. Similar to our baseline scenario, there is evidence of an uncertainty threshold effect in the Fama regression model. As before, the slope coefficient estimates are closer to one in the low uncertainty regime than in the high uncertainty regime. However, these estimates are statistically different from one in both uncertainty regimes, thus suggesting that the UIP condition does not hold even in the low uncertainty regime. Moreover, the estimated intercept parameters are not significantly different across regimes.

The low uncertainty results in the threshold regression model for longer time horizons are similar to those reported in Lee (2011) but in the context of the simple Fama model. There is evidence that the consensus forecast exchange rate for the USD/EUR rate during the 2002–2009 period was more precise in the three-month than in the 12-month horizon (Novotný & Raková, 2011). If this latter result were

¹⁷ For Mexico and India, we had to complete the monthly interest rate data obtained from Bloomberg and central banks by extrapolating the one-year interest rates for certificates of deposit.

Table 16 UIP *ex-post* panel data estimation

	Linear estimation	Threshold estimation			
		Country EPU	Anchored EPU	Weighted 1 EPU	Weighted 2 EPU
$\hat{\beta}$	0.659 (0.244)	—	—	—	—
$\hat{\alpha}_1 - \hat{\alpha}_2$	—	− 0.007 (0.013)	0.001 (0.008)	− 0.005 (0.008)	− 0.023*** (0.007)
<i>Low uncertainty</i>					
$\hat{\beta}_1$	—	0.808 (0.226)	1.228 (0.295)	1.082 (0.256)	1.286 (0.281)
<i>High uncertainty</i>					
$\hat{\beta}_2$	—	− 1.201*** (0.575)	− 0.119*** (0.307)	− 0.252*** (0.381)	− 0.233*** (0.323)
Threshold estimate	—	286.177	120.544	147.589	127.940
90% confidence interval	—	[62.4, 348.7]	[116.6, 125.9]	[131.2, 149.5]	[127.3, 145.2]
Test for threshold effects	—	0.483	0.079	0.096	0.098
Observations	3192	3192	3192	3192	3192
Countries	14	14	14	14	14
Period	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12

Heteroscedasticity and autocorrelation consistent (HAC) standard errors are in parentheses, lag length is set to $T^{\frac{1}{4}}$. The test for threshold effects shows the probability value for the null hypothesis of $\hat{\alpha}_1 = \hat{\alpha}_2$ and $\hat{\beta}_1 = \hat{\beta}_2$; we used 1000 bootstrap replications for the test

*, ** and *** denote statistical significance at the 10, 5 and 1% level, respectively, for the individual null hypotheses of the difference between regime intercepts equal to zero and a slope equal to 1

true for all the forecast exchange rates utilized in our study, this could partly explain the better findings obtained in our baseline scenario.

Maximum likelihood estimation

Are our main threshold panel regression findings robust to the estimation method? In this regard, Ramírez-Rondán (2020) proposes a maximum likelihood (ML) approach to estimate a dynamic panel threshold model. Although the model we are dealing with is not strictly dynamic, this is a particular case of the family of models that can be estimated with the ML method.

In order to eliminate the country-level fixed effect, we take the first difference in Eq. (5). This results in

$$\begin{aligned} \Delta(\hat{s}_{it+h} - s_{it}) = & (\alpha_1 - \alpha_2)\Delta 1(q_{it} \leq \gamma) + \beta_1 \Delta(i_{h,it} - i_{h,it})^+ \\ & + \beta_2 \Delta(i_{h,it} - i_{h,it})^\pm + \Delta e_{it+h}, \end{aligned} \quad (16)$$

Table 17 Panel data estimation with one-year forecast horizon

	Linear estimation	Threshold estimation			
		Country EPU	Anchored EPU	Weighted 1 EPU	Weighted 2 EPU
$\hat{\beta}$	0.358** (0.050)	—	—	—	—
$\hat{\alpha}_1 - \hat{\alpha}_2$	—	0.001 (0.002)	0.004 (0.001)	0.001 (0.001)	0.002 (0.001)
<i>Low uncertainty</i>					
$\hat{\beta}_1$	—	0.618*** (0.082)	0.593*** (0.061)	0.693*** (0.051)	0.643*** (0.052)
<i>High uncertainty</i>					
$\hat{\beta}_2$	—	0.138*** (0.059)	0.174*** (0.044)	0.170*** (0.048)	0.148*** (0.045)
Threshold estimate	—	120.478	126.626	120.676	127.940
90% confidence interval	—	[111.7, 127.8]	[126.6, 126.6]	[108.7, 124.7]	[100.4, 129.5]
Test for threshold effects	—	0.016	0.053	0.020	0.022
Observations	3192	3192	3192	3192	3192
Countries	14	14	14	14	14
Period	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12

Heteroscedasticity and autocorrelation consistent (HAC) standard errors are in parentheses, lag length is set to $T^{\frac{1}{4}}$. The test for threshold effects shows the probability value for the null hypothesis of $\hat{\alpha}_1 = \hat{\alpha}_2$ and $\hat{\beta}_1 = \hat{\beta}_2$; we used 1000 bootstrap replications for the test

*, ** and *** denote statistical significance at the 10, 5 and 1% level, respectively, for the individual null hypotheses of the difference between regime intercepts equal to zero and a slope equal to 1

where $\Delta(\hat{s}_{it+h} - s_{it}) = (\hat{s}_{it+h} - s_{it}) - (\hat{s}_{it-1+h} - s_{it-1})$, $\Delta 1(q_{it} \leq \gamma) = 1(q_{it} \leq \gamma) - 1(q_{it-1} \leq \gamma)$; $\Delta(i_{h,it} - i_{h,it})^+ = (i_{h,it} - i_{h,it})1(q_{it} \leq \gamma) - (i_{h,it-1} - i_{h,it-1})1(q_{it-1} \leq \gamma)$; $\Delta(i_{h,it} - i_{h,it})^\pm = (i_{h,it} - i_{h,it})1(q_{it} > \gamma) - (i_{h,it-1} - i_{h,it-1})1(q_{it-1} > \gamma)$; and $\Delta e_{it+h} = e_{it+h} - e_{it-1+h}$.¹⁸

Let the stacked data and errors for a country be noted as

$$X_i(\gamma) = \begin{bmatrix} \Delta 1(q_{i1} \leq \gamma) & \Delta(i_{h,i1} - i_{h,i1}^*)^+ & \Delta(i_{h,i1} - i_{h,i1}^*)^\pm \\ \vdots & \vdots & \vdots \\ \Delta 1(q_{iT} \leq \gamma) & \Delta(i_{h,iT} - i_{h,iT}^*)^+ & \Delta(i_{h,iT} - i_{h,iT}^*)^\pm \end{bmatrix};$$

¹⁸ Note that given the nature of the panel threshold model, α_1 and α_2 cannot be recovered directly; that is, since the first difference produces $\alpha_1(1(q_{it} \leq \gamma) - 1(q_{it-1} \leq \gamma)) + \alpha_2(1(q_{it} > \gamma) - 1(q_{it-1} > \gamma)) = (\alpha_1 - \alpha_2)(1(q_{it} \leq \gamma) - 1(q_{it-1} \leq \gamma))$, where we use that fact that $1(q_{it} > \gamma) = 1 - 1(q_{it} \leq \gamma)$.

Table 18 Maximum likelihood panel data estimation

	Linear estimation	Threshold estimation			
		Country EPU	Anchored EPU	Weighted 1 EPU	Weighted 2 EPU
$\hat{\beta}$	0.662*** (0.033)	—	—	—	—
$\hat{\alpha}_1 - \hat{\alpha}_2$	—	-0.007*** (0.002)	-0.002 (0.002)	-0.007** (0.003)	-0.008*** (0.002)
<i>Low uncertainty</i>					
$\hat{\beta}_1$	—	1.187* (0.096)	1.009 (0.082)	1.027 (0.092)	1.089 (0.113)
<i>High uncertainty</i>					
$\hat{\beta}_2$	—	0.046*** (0.080)	0.083*** (0.106)	-0.069*** (0.141)	0.043*** (0.113)
Threshold estimate	—	126.504	125.945	144.765	130.959
90% confidence interval	—	[112.9, 127.8]	[126.0, 130.5]	[116.6, 147.6]	[127.6, 133.8]
Test for threshold effects	—	0.000	0.064	0.007	0.003
Observations	3192	3192	3192	3192	3192
Countries	14	14	14	14	14
Period	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12	03m01–21m12

White standard errors are in parentheses. The test for threshold effects shows the probability value for the null hypothesis of $\hat{\alpha}_1 = \hat{\alpha}_2$ and $\hat{\beta}_1 = \hat{\beta}_2$; we used 1000 bootstrap replications for the test

*, ** and *** denote statistical significance at the 10, 5 and 1% level, respectively, for the individual null hypotheses of the difference between regime intercepts equal to zero and a slope equal to 1

$$Y_i = \begin{bmatrix} \Delta(\hat{s}_{i1+h} - s_{i1}) \\ \vdots \\ \Delta(\hat{s}_{iT+h} - s_{iT}) \end{bmatrix}; \text{ and } \Delta e_i = \begin{bmatrix} \Delta e_{i1+h} \\ \vdots \\ \Delta e_{iT+h} \end{bmatrix};$$

with this notation, the estimation procedure starts by fixing γ at any value in the empirical support of the threshold variable. Note that for any given γ , the maximum likelihood estimation is asymptotically equivalent to the minimum distance estimator $\sum_{i=1}^n \Delta e_i' \Omega^{-1} \Delta e_i$, where Ω is a matrix with values of twos and negative ones in the first and second main diagonals, respectively; and zeros otherwise.

Thus, after taking the first-order condition and by setting the partial derivative equal to zero, for any given γ , the regression coefficients $\alpha_1 - \alpha_2$, β_1 , and β_2 can be obtained by

$$\hat{\theta}(\gamma) = \left(\sum_{i=1}^n X_i(\gamma)' \Omega^{-1} X_i(\gamma) \right)^{-1} \left(\sum_{i=1}^n X_i(\gamma)' \Omega^{-1} Y_i \right), \quad (17)$$

where $\theta = (\alpha_1 - \alpha_2, \beta_1, \beta_2)'$, and then, the minimum distance estimator for a given threshold parameter γ is

$$\sum_{i=1}^n \Delta \hat{e}_i(\gamma)' \Omega^{-1} \Delta \hat{e}_i(\gamma). \quad (18)$$

where $\Delta \hat{e}_i(\gamma) = Y_i - X_i(\gamma)\hat{\theta}(\gamma)$.

The criterion function (18) is not smooth, since we previously estimated the threshold by using a grid search across the uncertainty space. Once $\hat{\gamma}$ is obtained, the regression coefficient estimates are then obtained as $\hat{\alpha}_1 - \hat{\alpha}_2 = \hat{\alpha}_1(\hat{\gamma}) - \hat{\alpha}_2(\hat{\gamma})$, $\hat{\beta}_1 = \hat{\beta}_1(\hat{\gamma})$, and $\hat{\beta}_2 = \hat{\beta}_2(\hat{\gamma})$. As regards the inference of the parameter estimates and testing for threshold effects, we follow the steps of the least-squares estimation of Hansen (1999). The difference in this case is that for the formulas $LR(\gamma)$ and F , $S(\gamma) = \sum_{i=1}^n \Delta \hat{e}_i(\gamma)' \Omega^{-1} \Delta \hat{e}_i(\gamma)$ is the minimum distance estimator, and $S_0 = \sum_{i=1}^n \Delta \hat{e}_i' \Omega^{-1} \Delta \hat{e}_i$ is the minimum distance estimator of the linear panel data model (13) in first differences.

In Table 18, we report the results of the minimum distance estimator, which are asymptotically equivalent to the maximum likelihood estimator. Overall, the main results are very similar to those reported in Table 6. Nevertheless, all slope estimates have slightly higher values, and one of these estimates is significantly different from one. Thus, we confirm the presence of threshold effects, the failure of UIP in the high uncertainty regime, and the general holding of UIP in the low uncertainty regime.

Acknowledgement We thank César Calderón, Luigi Donayre, Roberto Duncan, Jeff Frankel, Gaston Gelos, Anne Hansen, Matias Ossandon, Katheryn Niles Russ, the anonymous reviewers, and the participants of the 2022 CEMLA-FRBNY-ECB Conference, the Annual Event of Finance Research Letters 2022, the XXVI Meeting of the Central Bank Researchers Network, the 2020 European Winter Meeting of the Econometric Society, the 9th Annual Meeting of the Midwest Economics Research Group, the 2019 Annual Meeting of the Latin American and Caribbean Economic Association, the 2019 Congress of the Peruvian Economic Association, and the Research Seminars at the Universidad Diego Portales, Ohio University, the Universidad Autónoma Metropolitana, CIDE, and Fundação Getulio Vargas for useful comments. Andrea Vilchez, José Mendoza, Ronald Cueva, and Emiliano Rojas provided excellent research assistance at different project stages.

Declarations

Conflict of interest We declare that the resubmitted manuscript has no conflict of interest.

References

- Akram, Q. F., Rime, D., & Sarno, L. (2008). Arbitrage in the foreign exchange market: turning on the microscope. *Journal of International Economics*, 76(2), 237–253.
- Albagli, E., Ceballos, L., Claro, S., & Romero, D. (2023). *UIP: insights from event studies*. In Presented at the 2023 CEMLA Conference.
- André, M. C., Armijo, A., Medina Espidio, S., & Sandoval, J. (2023). Policy mix in a small open emerging economy with commodity prices. *Latin American Journal of Central Banking*, 4(1), 100082.

- Arroyo Marioli, F., Becerra, J. S., & Solorza, M. (2022). The credit channel in Chile through the lens of a semi-structural model. *Latin American Journal of Central Banking*, 3(2), 100056.
- Baillie, R. T., & Kilic, R. (2006). Do asymmetric and nonlinear adjustments explain the forward premium anomaly? *Journal of International Money and Finance*, 25(1), 22–47.
- Baker, D., Bloom, N., & Davis, S. (2016). Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4), 1593–1636.
- Balke, N. S., & Wohar, M. E. (1998). Nonlinear dynamics and covered interest rate parity. *Empirical Economics*, 23(4), 535–559.
- Batchelor, R. (2001). How useful are the forecasts of intergovernmental agencies? The IMF and OECD versus the consensus. *Applied Economics*, 33(2), 225–235.
- Berg, K., & Mark, N. (2018). Where's the risk? The forward premium bias, the carry-trade premium, and risk-reversals in general equilibrium. *Journal of International Money and Finance*, 95, 297–316.
- Berthold, B. (2023). The macroeconomic effects of uncertainty and risk aversion shocks. *European Economic Review*, 154, 104442.
- Bordo, M. D., Duca, J. V., & Koch, C. (2016). Economic policy uncertainty and the credit channel: Aggregate and bank level U.S. evidence over several decades. *Journal of Financial Stability*, 26, 90–106.
- Burnside, C., Eichenbaum, M., Kleshchelski, I., & Rebelo, S. (2006). *The returns to currency speculation*. NBER Working Papers 12489.
- Bussière, M., Chinn, M., Ferrara, L., & Heipertz, J. (2022). The new Fama puzzle. *IMF Economic Review*, 70, 451–486.
- Cerutti, E. M., Obstfeld, M., & Zhou, H. (2021). Covered interest parity deviations: Macrofinancial determinants. *Journal of International Economics*, 30, 103447.
- Chinn, M. D., & Frankel, J. (2019). *A third of a century of currency expectations data: The carry trade and the risk premium*. Mimeo.
- Chinn, M. D., & Meredith, G. (2004). Monetary policy and long horizon uncovered interest rate. *IMF Staff Papers*, 51(3), 409–430.
- Chinn, M. D., & Quayyum, S. (2013). *Long horizon uncovered interest parity re-assessed*. Mimeo, University of Wisconsin-Madison.
- Colacito, R., Riddiough, S., & Sarno, L. (2020). Business cycles and currency returns. *Journal of Financial Economics*, 137(3), 659–678.
- Corsetti, G., & Marin, E.A. (2020). *A century of arbitrage and disaster risk pricing in the foreign exchange market*. CEPR Discussion Papers 14497.
- Donayre, L. (2014). Estimated thresholds in the response of output to monetary policy: Are large policy changes less effective? *Macroeconomic Dynamics*, 18(1), 41–64.
- Du, W., Tepper, A., & Verdelhan, A. (2018). Deviations from covered interest rate parity. *Journal of Finance*, 73(3), 915–957.
- Engel, C. (1996). The forward discount anomaly and the risk premium: A survey of recent evidence. *Journal of Empirical Finance*, 3(2), 123–192.
- Engel, C. (2016). Exchange rates, interest rates, and the risk premium. *American Economic Review*, 106(2), 436–474.
- Engel, C., Kazakova, E., Wang, M., & Xiang, N. (2022). A reconsideration of the failure of uncovered interest parity for the U.S. dollar. *Journal of International Economics*, 136, 103602.
- Fama, E. (1984). Forward and spot exchange rates. *Journal of Monetary Economics*, 14(3), 319–338.
- Farhi, E., & Gabaix, X. (2016). Rare disasters and exchange rates. *Quarterly Journal of Economics*, 131(1), 1–52.
- Farhi, E., & Werning, I. (2014). Dilemma not trilemma? Capital controls and exchange rates with volatile capital flows. *IMF Economic Review*, 62(4), 569–605.
- Fernández, A., Klein, M. W., Rebucci, A., Schindler, M., & Uribe, M. (2016). Capital control measures: A new dataset. *IMF Economic Review*, 64(3), 548–574.
- Flood, R. P., & Rose, A. K. (2002). Uncovered interest parity in crisis. *IMF Staff Papers*, 49(2), 252–266.
- Froot, K. A., & Thaler, R. H. (1990). Anomalies: Foreign exchange. *Journal of Economic Perspectives*, 4(3), 179–192.
- Gruss, B., & Kebhaj, S. (2019). *Commodity terms of trade: A new database*. IMF Working Paper 19/21.
- Hansen, B. E. (1999). Threshold effects in non-dynamic panels: Estimation, testing and inference. *Journal of Econometrics*, 93(2), 345–368.
- Hansen, B. E. (2000). Sample splitting and threshold estimation. *Econometrica*, 68(3), 575–603.

- Herger, N. (2016). Panel data models and the uncovered interest parity condition: The role of two-way unobserved components. *International Journal of Finance and Economics*, 21(3), 294–310.
- Hollifield, B., & Uppal, R. (1997). An examination of uncovered interest rate parity in segmented international commodity markets. *Journal of Finance*, 52(5), 2145–2170.
- Husted, L., Rogers, J., & Sun, B. (2018). Uncertainty, currency excess returns, and risk reversals. *Journal of International Money and Finance*, 88, 228–241.
- Ilzetzki, E., Reinhart, C., & Rogoff, K. (2019). Exchange arrangements entering the 21st century: Which anchor will hold? *Quarterly Journal of Economics*, 134(2), 599–646.
- Ismailov, A., & Rossi, B. (2018). Uncertainty and deviations from uncovered interest rate parity. *Journal of International Money and Finance*, 88, 242–259.
- Lee, B.-J. (2011). Uncovered interest parity: Cross-sectional evidence. *Review of International Economics*, 19(2), 219–231.
- Lustig, H., Roussanov, N., & Verdelhan, A. (2011). Common risk factors in currency markets. *Review of Financial Studies*, 24(11), 3731–3777.
- Lustig, H., Stathopoulos, A., & Verdelhan, A. (2019). The term structure of currency carry trade risk premia. *American Economic Review*, 109(12), 4142–4177.
- Lloyd, S.P., & Marin, E.A. (2020). *Exchange rate risk and business cycles*. Staff Working Paper 872, Bank of England.
- Meese, R., & Rogoff, K. (1983). Empirical exchange rate models of the seventies, Do they fit out of sample? *Journal of International Economics*, 14(1–2), 3–24.
- Novotný, F., & Raková, M. (2011). Assessment of consensus forecasts accuracy: The Czech National Bank perspective. *Czech Journal of Economics and Finance*, 61(4), 348–366.
- Peel, D., & Taylor, M. P. (2002). Covered interest rate arbitrage in the interwar period and the Keynes–Einzig conjecture. *Journal of Money, Credit and Banking*, 34(1), 51–75.
- Ramírez-Rondán, N. R. (2020). Maximum likelihood estimation of dynamic panel threshold models. *Econometric Reviews*, 39(3), 260–276.
- Rehse, D., Riordan, R., Rottke, N., & Zietz, J. (2019). The effects of uncertainty on market liquidity: Evidence from Hurricane Sandy. *Journal of Financial Economics*, 134(2), 318–332.
- Sarno, L., Valente, G., & Leon, H. (2006). Nonlinearity in deviations from uncovered interest parity: An explanation of the forward bias puzzle. *Review of Finance*, 10(3), 443–482.
- Stavrakeva, V., & Tang, J. (2015). *Exchange rates and monetary policy*. Working Papers 15-16, Federal Reserve Bank of Boston.
- Verdelhan, A. (2010). A habit-based explanation of the exchange rate risk premium. *Journal of Finance*, 65(1), 123–146.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.